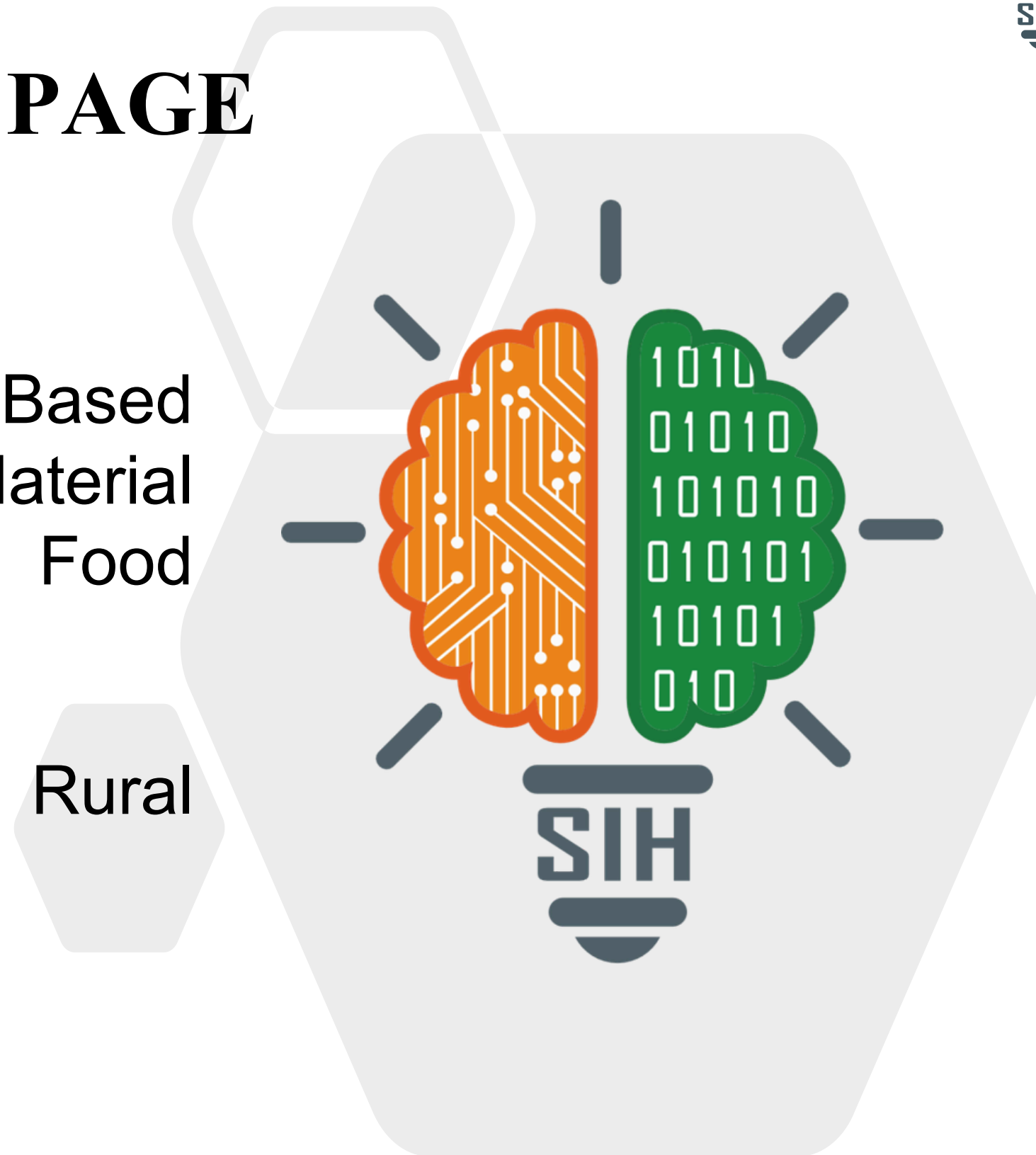


TITLE PAGE

- **Problem Statement ID-** SIH26236
- **Problem Statement Title-** AI-Based Intelligent Food Packaging Material Recommendation System for Food Commodities
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-**
- **Team Name-** Astra Squad



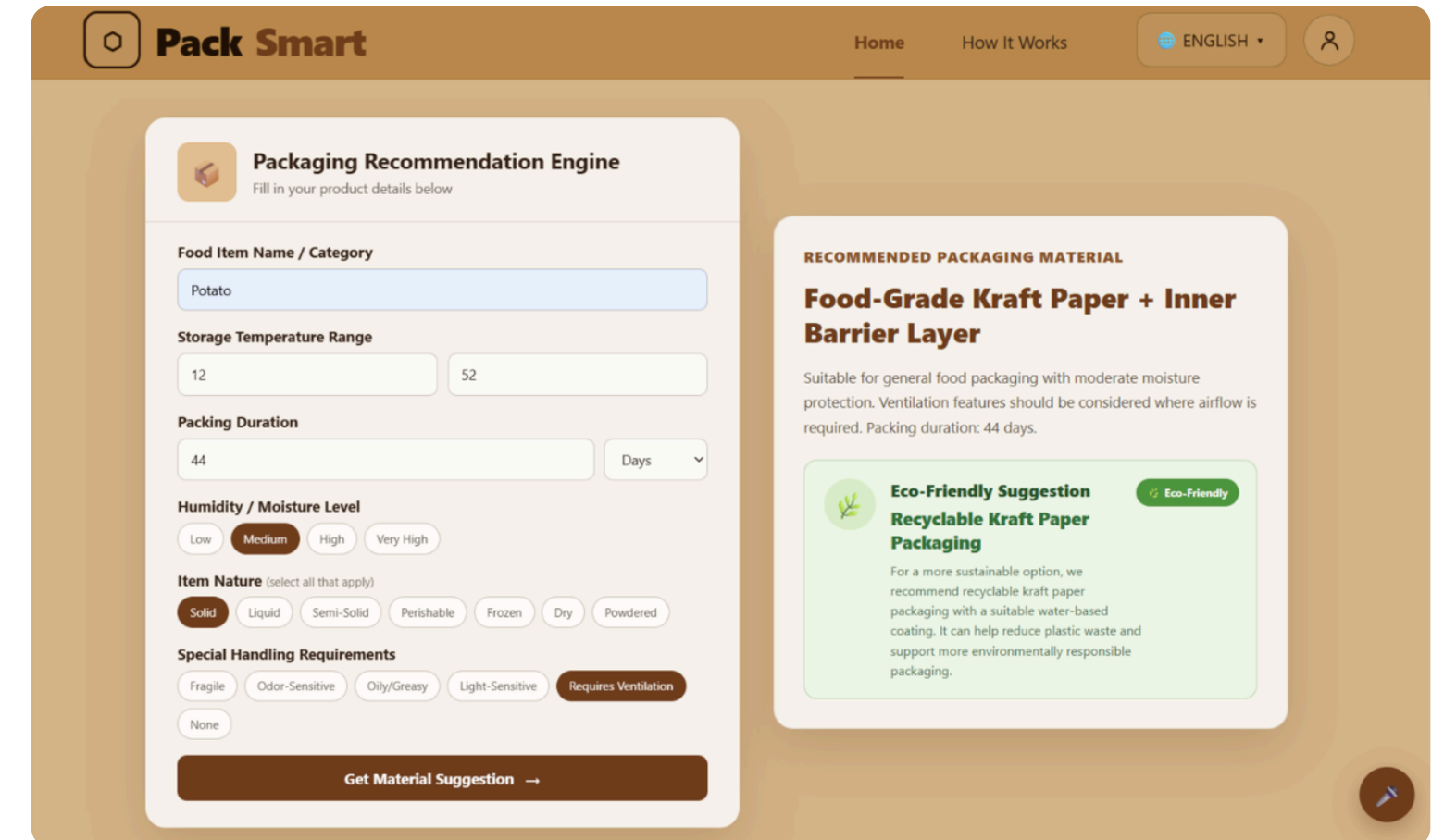
SMARTPACK : AI-POWERED PACKAGING RECOMMENDATION APPLICATION

PROPOSED SOLUTION

- **AI-Powered Recommendation:** Analyzes food properties and storage conditions to identify suitable packaging requirements.
- **Food-Packaging KB:** Combines food characteristics with packaging properties such as OTR, WVTR, strength, permeability, etc
- **Smart Recommendation Engine:** Review, Ranks, Recommend suitable materials using ML, domain rules and compatibility scoring, with clear stats, advantages and Key features.

HOW IT ADDRESSES THE PROBLEM

- **Understands Food Requirements:** Converts food & storage conditions into specific packaging requirements.
- **Evaluates Multiple Materials:** Compares options such as LDPE, HDPE, PET, metallized and biodegradable materials.
- **Explains Each Recommendation:** Shows suitability scores, key properties and clear reasons behind the recommended materials.



The image shows a web application prototype for 'Pack Smart'. The header includes a home icon, the title 'Pack Smart', and navigation links for 'Home' and 'How It Works'. There is a language selector set to 'ENGLISH' and a user profile icon. The main content area is titled 'Packaging Recommendation Engine' with a subtext 'Fill in your product details below'. It contains several input fields: 'Food Item Name / Category' (with 'Potato' entered), 'Storage Temperature Range' (with '12' and '52' entered), 'Packing Duration' (with '44' entered and a 'Days' dropdown), 'Humidity / Moisture Level' (with 'Medium' selected), and 'Item Nature' (with 'Solid' selected). There are also 'Special Handling Requirements' checkboxes for 'Fragile', 'Odor-Sensitive', 'Oily/Greasy', 'Light-Sensitive', 'Requires Ventilation', and 'None'. A 'Get Material Suggestion' button is at the bottom. On the right, a 'RECOMMENDED PACKAGING MATERIAL' section displays 'Food-Grade Kraft Paper + Inner Barrier Layer' as the primary recommendation, with a description of its suitability. Below this, an 'Eco-Friendly Suggestion' box recommends 'Recyclable Kraft Paper Packaging' with a brief explanation of its benefits. An 'Eco-Friendly' badge is also present.

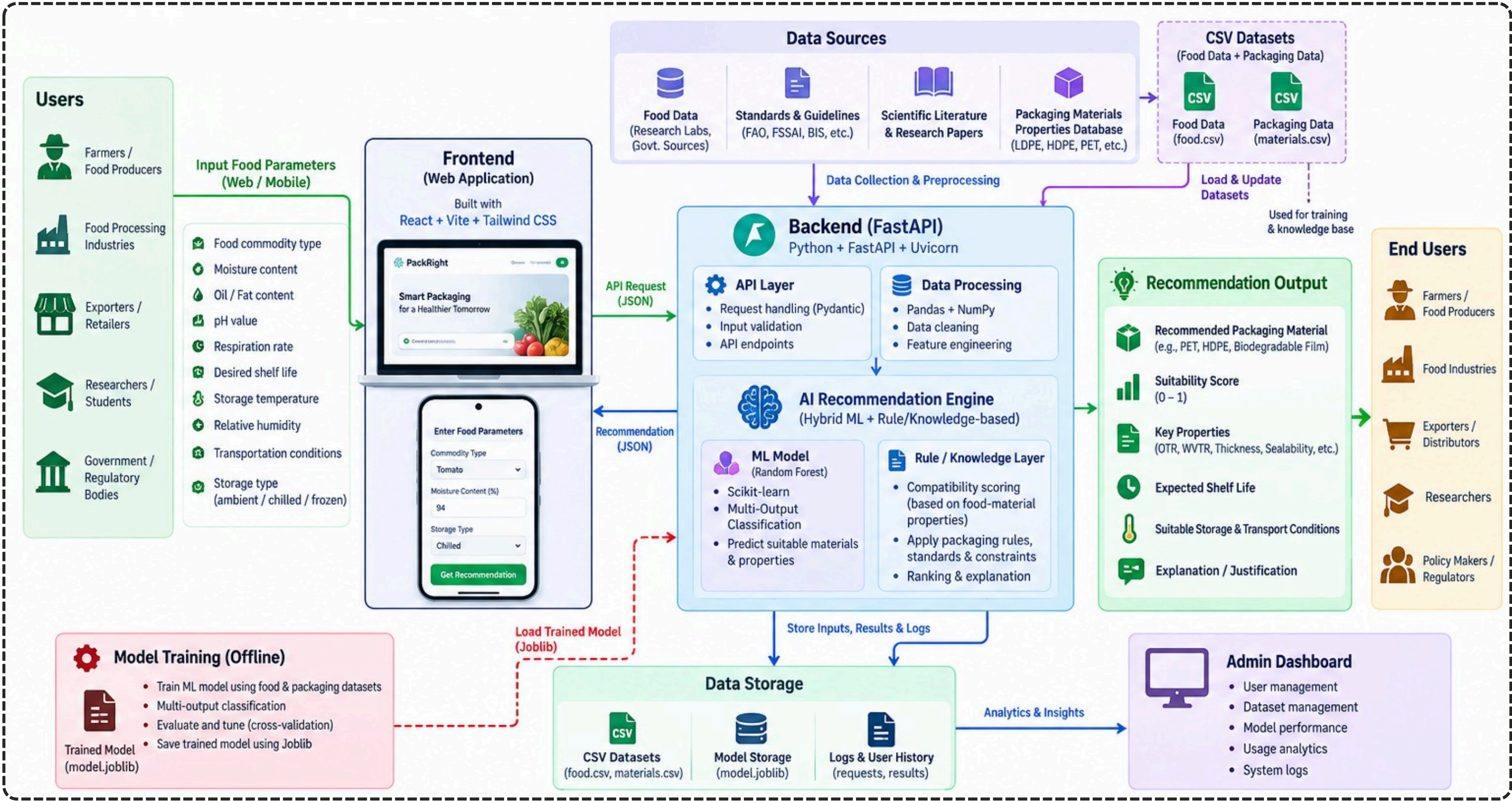
(SMARTPACK APPLICATION PROTOTYPE)

INNOVATION & UNIQUENESS

Beyond Material Lookup, Hybrid Intelligence, Performance-Driven Selection, Explainable Choices

SYSTEM ARCHITECTURE & DATA FLOW DIAGRAM

TECH STACK



CHALLENGES & RISKS

- **Cost:** High-performance packaging may be costly for MSMEs & farmers.
- **Data Gaps:** Limited data for less-studied commodities.
- **Regional Variability:** Climate, storage & transport can affect performance.
- **Compliance & Adoption:** Regulatory trade-offs and unclear ROI may slow adoption.

MITIGATION STRATEGIES

- **Cost Flexibility:** Offer budget, standard & premium packaging options.
- **Data & Accuracy:** Build an evidence-based knowledge base with continuous updates.
- **Regional Adaptation:** Factor in climate, storage & supply-chain conditions.
- **Sustainable & Validated:** Prioritize recyclable options and validate through FPO/MSME pilots.

FEASIBILITY

- **Proven Materials:** Builds on commercially available MAP, barrier, active/breathable and bio-based packaging.
- **Available Data:** Uses existing food properties and packaging parameters such as OTR, WVTR, strength and sealability.
- **Practical AI:** Combines ML prediction, domain rules and compatibility scoring to rank suitable materials.
- **Scalable MVP:** Starts with selected commodities and materials, with scope to expand the knowledge base and ML model.

POTENTIAL IMPACT ON TARGET AUDIENCE

- **Farmers & FPOs:** Reduce spoilage across 68.90 MT estimated losses.
- **MSMEs & Processors:** Lower ₹1.49 lakh crore economic losses through better packaging selection.
- **Startups & Exporters:** Improve shelf life & transport protection with data-driven recommendations.
- **Packaging Technologists:** Compare barrier, shelf-life & sustainability factors in one system.

Benefits of the Solution

- **Economic:** Reduce 68.90 MT losses & ₹1.49 lakh crore economic loss.
- **Social:** Support Farmers, FPOs & MSMEs with accessible packaging guidance.
- **Food Quality:** Improve shelf life & product protection.
- **Environmental:** Reduce food waste linked to 8–10% of global GHG emissions.
- **Supply Chain:** Reduce damage & spoilage during storage and transport.
- **Technical:** Enable data-driven, explainable material selection

Key References

- [1] [Ministry of Food Processing Industries \(MoFPI\), Technical Reports on Harvest and Post-Harvest Losses of Major Crops and Commodities in India.](#)
- [2] [Food Safety and Standards Authority of India \(FSSAI\), Food Safety and Standards \(Packaging\) Regulations, 2018.](#)
- [3] [United Nations Environment Programme \(UNEP\), Food Waste Index Report 2024.](#)
- [4] [MoFPI, Study to Determine Post-Harvest Losses of Agri Produce in India, 2025.](#)
- [5] [Sagar, Narashans & Rani, Nitu. \(2026\). Recent trends and innovations in smart and AI-based food packaging: A review. Frontiers in Food Science and Technology. 5. 10.3389/frfst.2025.1665055.](#)